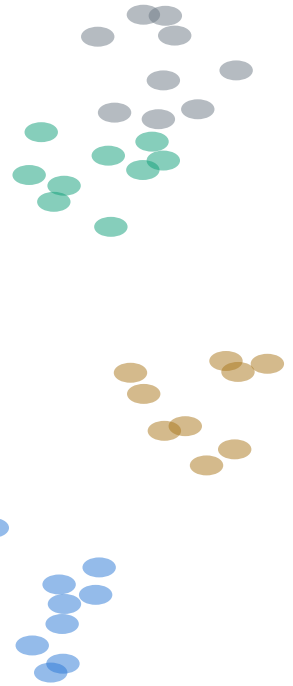
HEPTOKENS · CAMPAIGN 6 · REAL CMS DATA (AOJ)

Beyond the VQ-VAE



Alternate tokenization schemes, put to the test — rate-matched VAE frontier · physics grid · FSQ.

23 runs · headline arms × 3 seeds · pre-registered outcomes

WHERE CAMPAIGNS 1–5 LEFT US

Complementary tokens, one open corner

01 **Library owns the criterion**
recall@8 0.9991 vs matched VQ 0.927 — ~80× fewer retrieval misses.

02 **VQ owns the information**
within-PID pT R^2 0.97, PID/charge linearly decodable, 27–30 bits.

03 **Placement decides**
physics kernel on a smooth latent works (0.38→0.92); through a codebook it hurts (0.945→0.827).

04 **Two robust negatives**
criterion-shaped VQ codebooks; jet context at low rate.

05 **The open corner**
no trained token is sub-raw AND faithful ($R^2 \geq 0.9$) AND criterion-perfect (recall≥0.999).

THE EVALUATION

Survey everything, rank the frontier, test the cheapest

Full-repo audit + 2023–26 literature scan across three families, then an adversarial pass over every 'settled' claim. Campaign 6 = the two sanctioned steps + the one new family that could still deliver a sub-raw criterion-strong discrete token.

DISCRETE

FSQ / BSQ (codebook-free)
→ C6c

physics-grid binning
→ C6b

rotation-trick · SimVQ retest
queued

Gumbel / dVAE
queued

CONTINUOUS

β-sweep: rate-matched VAE
→ C6a

REPA frozen-teacher distill
queued

invertible flow tokens
queued

pseudo-2nd-detector proto
queued

RELEASE / CONSUMER

masked-continuous FM probe
→ C7

PID⊕charge symbols ⊕ latent
queued

entropy codec for μ tokens
queued

jet-level side token
queued

CAMPAIGN 6 · DESIGN

Three arms, outcomes pre-registered

6A · RATE-MATCHED VAE

β × dim × free-bits sweep, kernel on μ

pre-registered: retire VQ if recall ≥ 0.945 AND R^2 ≥ 0.97 at ≤ 30 bits — else map the saturation

6B · PHYSICS GRID

UNTRAINED: ln(1.1) log-pT, 0.1 η/φ bins, exact PID/charge (+½-bin offset copy)

pre-registered: any learned discrete token below this curve is retired

6C · FSQ

codebook-free quantizer, zero params; kernel on pre-rounding z_e (z_q + recon ablations)

pre-registered: sub-raw criterion token — or upgrade the negative to 'quantization per se'

6A · RATE-MATCHED VAE · HEADLINE

Criterion parity at ~13 KL-bits — but VQ is not retired

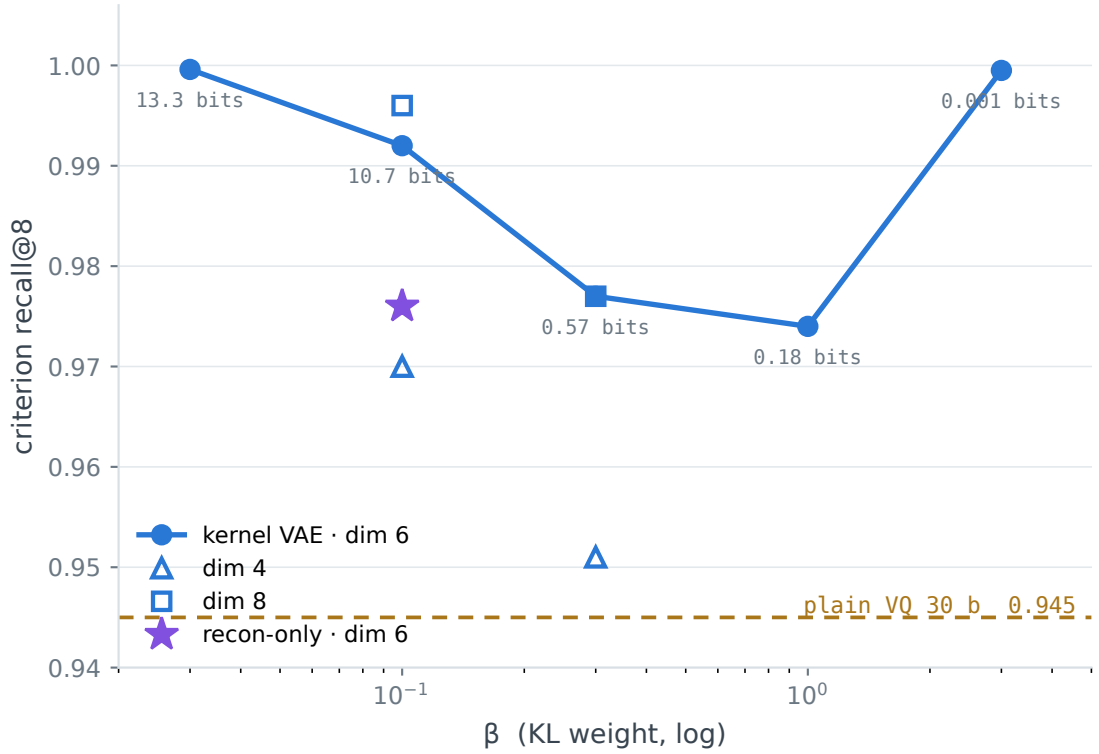
0.9984 ± 0.0015

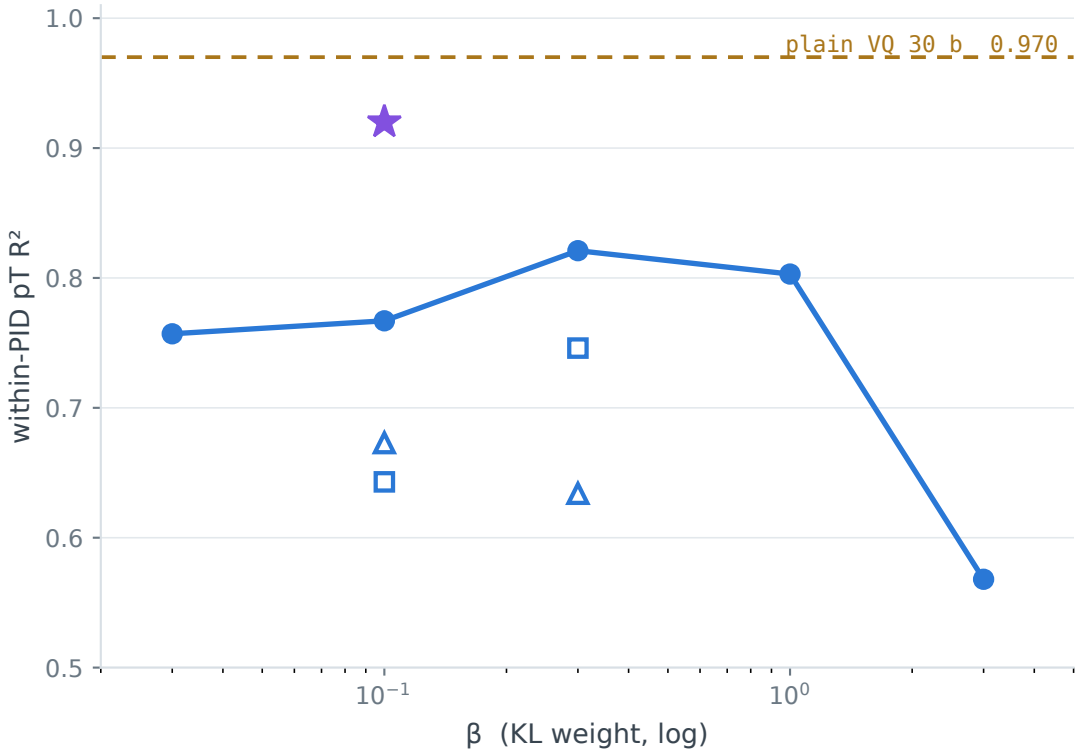
criterion recall@8 · β = 0.03, dim 6 · 3 seeds · ~13 KL-bits

Parity with the 96-bit deterministic dim-6 library, and far above plain VQ (0.945 at 30 bits).
But pT R^2 tops out at ≈ 0.82 anywhere on the sweep — short of the 0.97 retirement bar.
Organization-vs-information complementarity survives rate-matching.

6A · THE B DIAL

β trades information, not the criterion





Recall stays ≥ 0.95 across the dial (labels: measured KL-rate); R² peaks ≈ 0.82 at β = 0.3 — never reaching VQ. Recon-only star: 0.976 / 0.920.

6A · AN ACCOUNTING ARTIFACT, MEASURED

KL-rate is not a release rate for deterministic-μ tokens

recall 0.9995 @ 0.001 KL-bits

At β = 3 the posterior sits essentially on the prior — measured KL ≈ 0 — yet the released posterior mean still carries the full criterion geometry, because cosine ranking is scale-invariant: a tiny-norm but structured μ ranks partners perfectly.

Consequence: Campaign 5's celebrated '~1.4-bit' contrastive VAE was this artifact in milder form. Until an operational entropy codec exists, continuous-token rates are quoted as fp16 dim×16; KL-bits are diagnostics only.

6A · A C5 CLAIM, REVISED

'Reconstruction buys species, not the criterion' — a rate artifact

RECON-ONLY VAE · B=1 (C5)

recall 0.383

- the 2.5-bit bottleneck was the cause

RECON-ONLY VAE · B=0.1 (C6)

recall 0.976 ± 0.004

- pT R^2 0.920 · PID/charge probes 1.0
- ~11 KL-bits · 3 seeds

But it still does not beat the untrained null (0.983 / 0.949) on either axis — the physics kernel remains the only objective that ever has (0.9984 vs 0.9833).

6B · THE UNTRAINED REFERENCE BITES

Zero parameters ≈ trained VQ, at fewer bits

PHYSICS GRID, DOUBLE

0.9655 @ 26.3 bits

- pT R^2 0.949 · probes 1.0
- computable on ANY detector
- ln(1.1) pT · 0.1 η/φ · exact PID

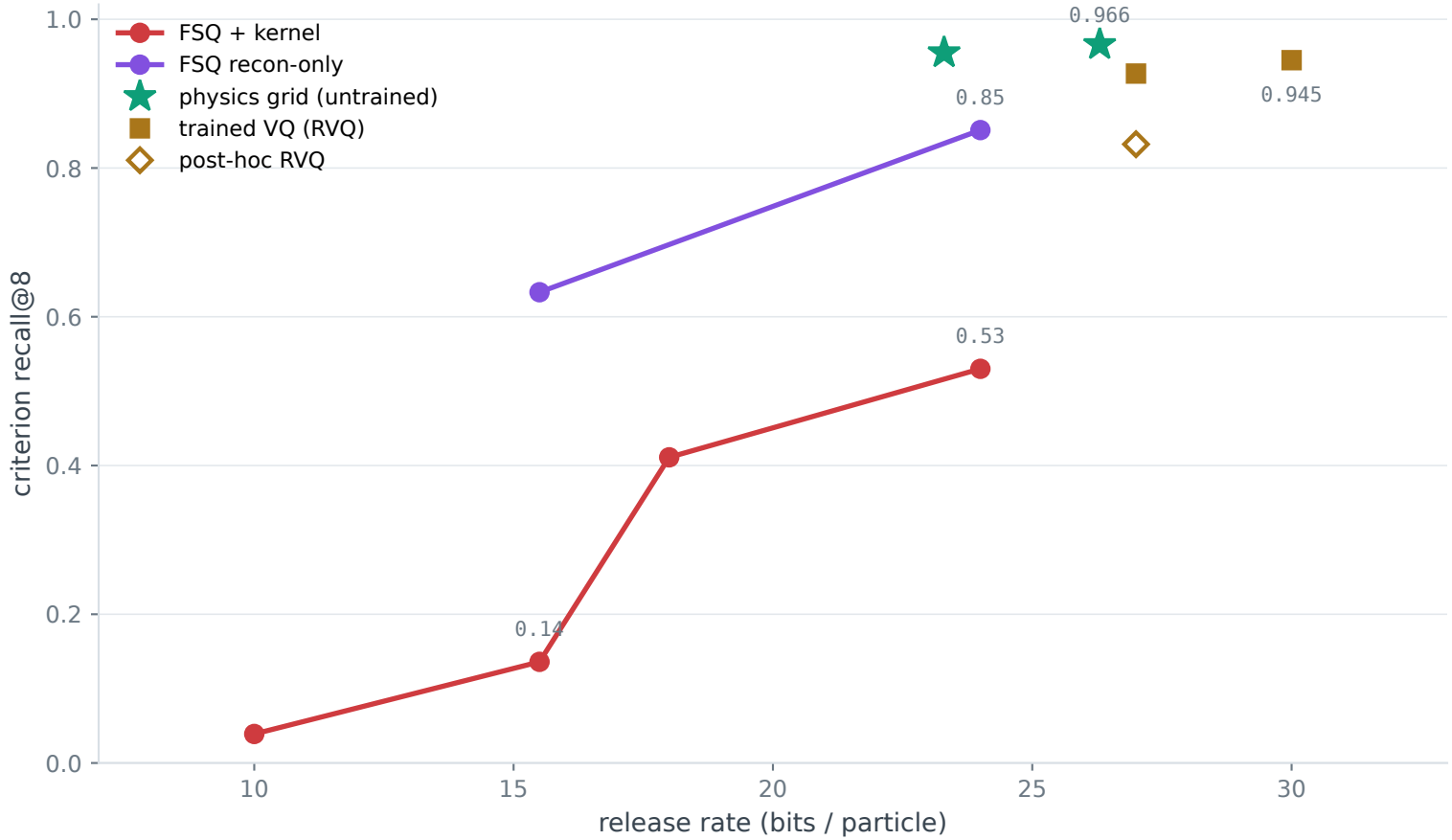
PLAIN VQ 3×1024 (TRAINED)

0.945 @ 30 bits

- pT R^2 0.970 · probes 1.0
- surviving edge: +0.02 R^2
- purity +0.017

At the 23–30-bit tier, learning a discrete token currently buys almost nothing over criterion-aligned binning — and the grid is a natural cross-experiment release tier.

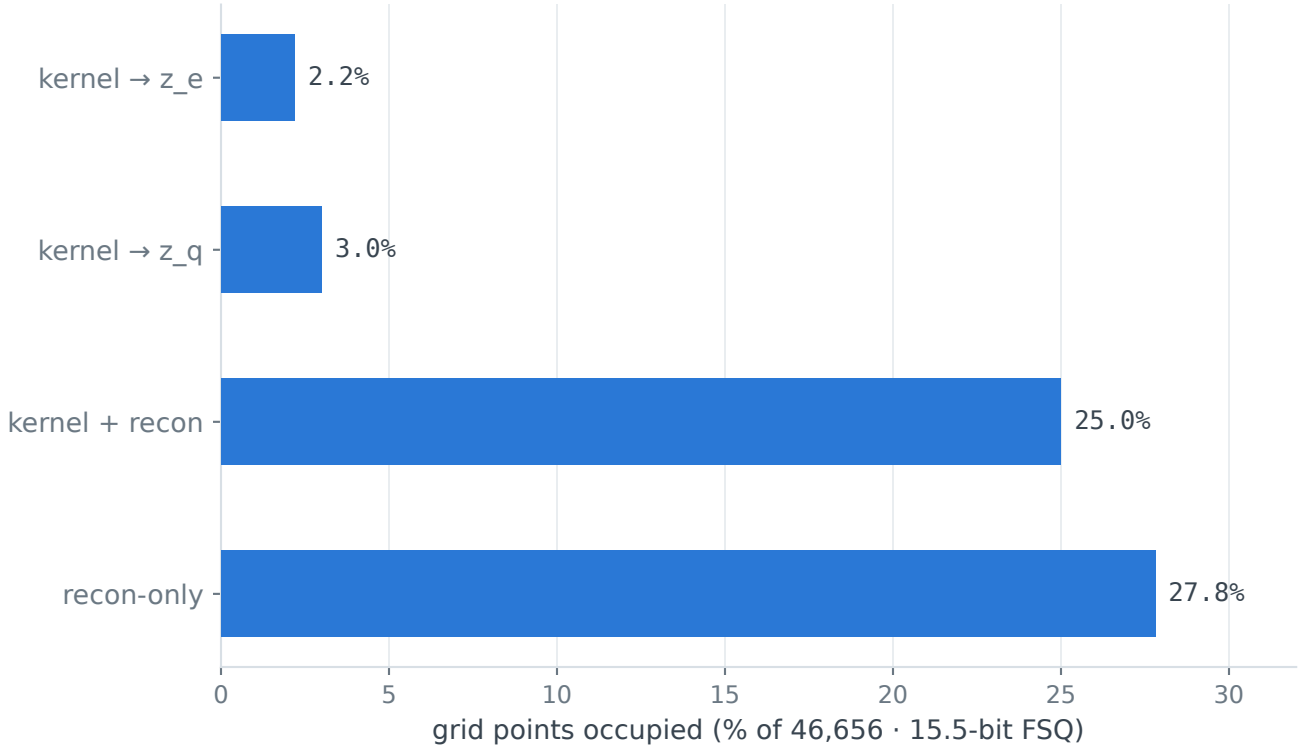
FSQ: the kernel hurts at every rate



No codebook, no EMA, no commitment loss — and criterion-shaping still loses to plain reconstruction at 10, 15.5, 18 and 24 bits. Both lose to the untrained grid. The C4/C5 negative was never about codebook dynamics.

6C · THE MECHANISM, CAUGHT ON CAMERA

The kernel collapses the code space

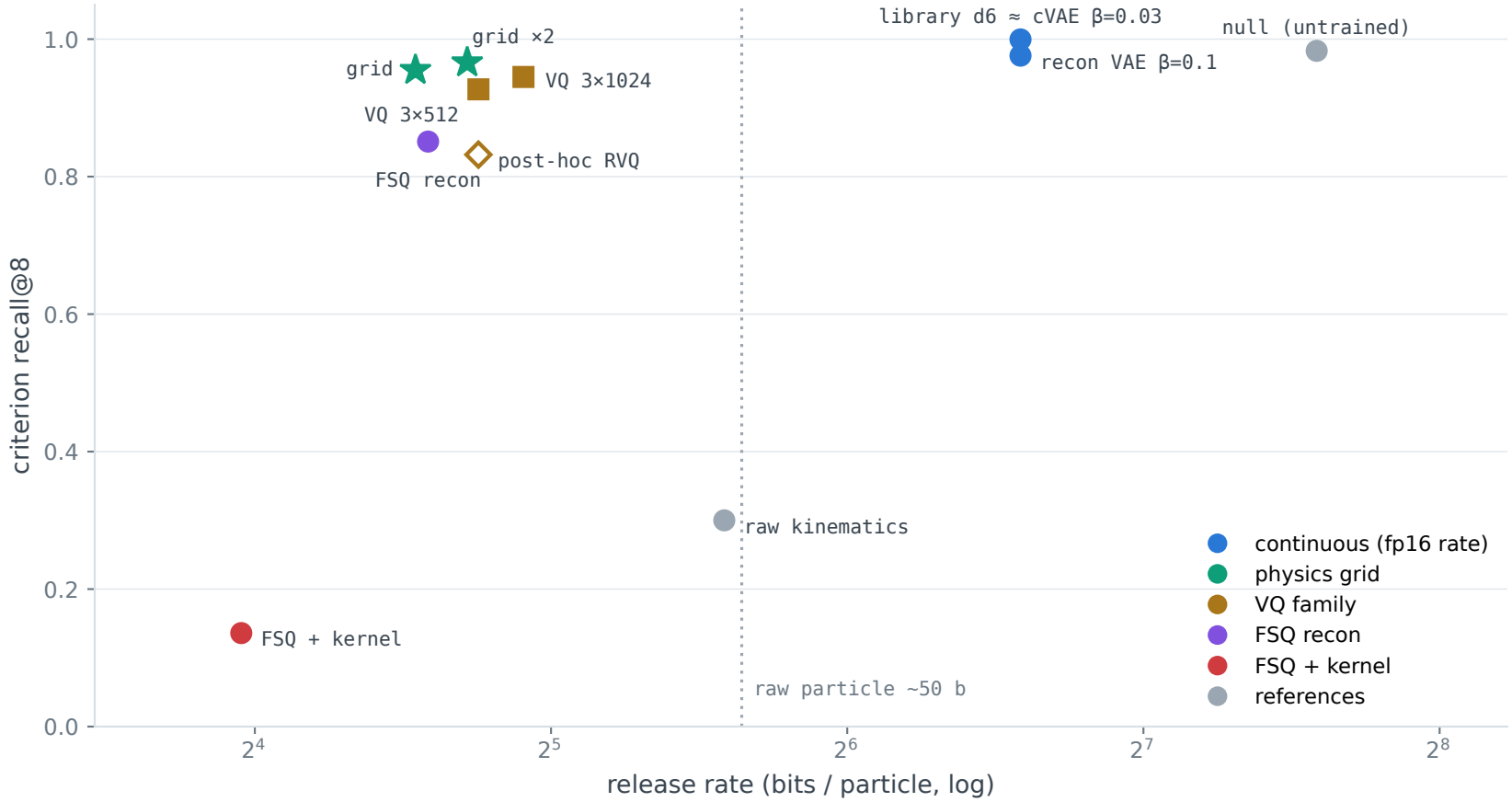


The kernel's fixed point drives criterion partners toward coincidence. Any quantizer turns coincidence into code degeneracy — thousands of particles share one grid point — and retrieval cannot rank true partners among identical-code impostors.

Verdict: install the criterion on a continuous latent; if a discrete tier is needed, bin the features (6b), not the embedding.

EVERYTHING ON ONE AXIS

The decision map, at honest rates



Continuous tokens plotted at their fp16 release size per slide 7 (the β=0.03 arm's 13.3 KL-bits is a diagnostic, not a rate). Full map incl. every C1–6 arm: benchmarks/figures/library_rate.png.

THE CLAIMS LEDGER

What Campaign 6 changed

OVERTURNED	**'recon buys species, not the criterion'** β=1 rate artifact — 0.976 at β=0.1.
RELABELED	**the '~1.4-bit' contrastive VAE** KL-rate ≠ release rate for deterministic μ; quote fp16.
CLOSED	**criterion-shaped discrete tokens** kernel hurts every quantizer at every rate; code degeneracy.
CONFIRMED	**complementarity (geometry vs information)** survives rate-matching: recall 0.998 @ 13 KL-bits, $R^2 \leq 0.82$.
NEW BAR	**untrained physics grid** 0.966 @ 26.3 bits — every discrete proposal must beat it.

KEY TAKEAWAYS

The release menu, as it stands

CRITERION + COMPACT

contrastive VAE

- recall 0.998 · dim 6
- 96 fp16 bits (honest rate)

DISCRETE TIER

physics grid ×2

- 0.966 @ 26.3 bits · R^2 0.95
- zero params · any detector

MAX FIDELITY

plain VQ 30 b

- R^2 0.970 · probes 1.0
- only if +0.02 R^2 matters

The corner (sub-raw ∧ faithful ∧ criterion-perfect) is still open — and now provably not reachable by quantizing the criterion.

NEXT

Campaign 7: let the consumer decide

RUNS TODAY

Masked-continuous FM probe over the release menu

Pretrain a backbone on each frozen export — no Campaign 3–6 winner has a consumer number yet.

THE OPEN CORNER

REPA-style distillation · PID⊕charge symbols ⊕ kinematic latent

Mechanism-changed attacks on recall ≥ 0.999 ∧ R^2 ≥ 0.9 — plus d0/dz restoration and the anomaly probe.

North star unchanged and still data-blocked: cross-experiment invariance — the grid tier and the context-free library are both built for it.